

Sarah M. Coleman, PhD

Open to relocation nationwide · Available to start within 30 days, at own expense

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Summary

Bioprocess development engineer specializing in **upstream process development** and **fed-batch fermentation**. Designed and implemented a first-in-company **fed-batch bioreactor** process end to end, covering feed strategy, control parameters, **SOPs**, **batch records**, and KPIs, and led **tech transfer** to manufacturing. Background spans microbial strain and fermentation process development, multi-omics analysis, and statistical modeling, with upstream PD experience at **Bristol Myers Squibb** (biologics, **CHO cell culture**) and **Takeda** (vaccines, Vero cells). PhD in Chemical Engineering and MS in Statistics, with experience in **scale-up**, **DoE**, and statistical analysis of experimental biology data in Python and R.

Work Experience

BioReNuva, Austin, TX

Bioprocess Development Engineer (Contract/Fixed-term) | Jun 2026 – Present

- Established fed-batch fermentation capability for glycolipid production using *Pseudomonas* strains as the company's subject-matter expert, building on prior collaboration with the Alper Lab
- Designed and implemented the company's first fed-batch bioreactor process from scratch, including feed strategy, control parameters, and run protocols
- Authored the standard operating procedure (SOP) for bioreactor runs, establishing the reference process documentation used for ongoing and future fermentation campaigns
- Defined key performance indicators (KPIs) and identified process bottlenecks for glycolipid synthesis, informing the roadmap for scale-up and continuous improvement
- Led cross-functional technology transfer, translating process knowledge into documentation and protocols to hand off from development to manufacturing teams

The University of Texas at Austin, Austin, TX

Postdoctoral Research Fellow, Statistics and Data Sciences | Oct 2024 – Sep 2026

- Developed epidemiological and statistical models of environmental-health data in Dr. Roger D. Peng's group, including analyses across 5 Texas cities and dimension reduction of emission mixtures across approximately 3,000 U.S. counties, resulting in a first-author paper in *Environmental Research Communications* and a second under review at *Environmental Health*; advised medical faculty on statistical methodology and experimental design

The University of Texas at Austin, Austin, TX

Lecturer, Statistics and Data Sciences | Jan 2026 – Apr 2026

- Instructor of record for 100-student upper-division data science and machine learning course, rated 4.32/5 overall by the students, a higher rating than the department, college, and university averages; managed 3 TAs

The University of Texas at Austin, Austin, TX

Graduate Student Fellow, Chemical Engineering | Aug 2019 – Aug 2024

- Integrated CFD modeling with fermentation experiments to characterize bioreactor performance at up to 50% (v/v) waste cooking oil in 3 L bioreactors, among the highest loadings reported, and simulated scale-up to 4,000 L
- Evolved *Yarrowia lipolytica* to tolerate up to 30% (v/v) hydrothermal liquefaction wastewater (up from 10%), yielding an isolate with nearly 3-fold higher titer of the high-value polyketide triacetic acid lactone while reducing COD (chemical oxygen demand)
- Evolved *Y. lipolytica* for growth on high-density waste cooking oil and engineered it to produce long-chain fatty alcohols, characterizing the evolved strain by genomics, transcriptomics, and metabolomics
- Developed a genome editing system using Golden Gate assembly for targeted engineering of *Y. lipolytica*, and used this system to screen a native promoter library
- Proposed and validated a linear mixed-effects model to quantitatively evaluate the performance of biological Boolean logic gates for the first time, published in *ACS Synthetic Biology*
- Mentored 4 undergraduate researchers and 2 junior PhD students, and presented results to project sponsors and collaborators

MathWorks, Remote (Natick, MA)

MATLAB Student Ambassador | Mar 2020 – Jul 2021

- Taught technical workshops on MATLAB-based modeling and data analysis workflows for academic researchers

Massachusetts Institute of Technology, Cambridge, MA

Advanced Undergraduate Researcher | Sep 2018 – Aug 2019

- Developed a method of nanophosphor size sorting in Dr. Michael S. Strano's lab to improve the capacitance of nanoparticles infiltrated into a light emitting plant

Massachusetts Institute of Technology, Cambridge, MA

Undergraduate Researcher | Sep 2016 – Aug 2018

- Developed anaerobic assays in Dr. Gregory Stephanopoulos's lab to characterize the carbon assimilation of *Moorella thermoacetica* enzymes in the Wood-Ljungdahl pathway

Bristol Myers Squibb, Devens, MA

Intern, Upstream Process Development (Biologics) | Jun 2018 – Aug 2018

- Designed and executed metabolite-controlled fed-batch cell culture experiments in 12 parallel 5 L bioreactors to systematically evaluate monoclonal antibody production performance in CHO cell lines
- Analyzed upstream bioprocess data to identify conditions that improved yield stability and process robustness in biologics production

Takeda Pharmaceuticals, Cambridge, MA

Intern, Process Development (Vaccines) | May 2017 – Aug 2017

- Optimized Vero cell culture conditions to improve vaccine production performance within a next-generation manufacturing system
- Ran tangential flow filtration (TFF) for vaccine manufacturing, and helped design experiments to improve process scalability and production efficiency

Education

The University of Texas at Austin, Austin, TX

Ph.D. in Chemical Engineering, Aug 2024 (Advisor: Dr. Hal S. Alper)

The University of Texas at Austin, Austin, TX

M.S. in Statistics, May 2024

Massachusetts Institute of Technology, Cambridge, MA

S.B. in Chemical-Biological Engineering, Jun 2019 (Research advisors: Dr. Michael S. Strano, Dr. Gregory Stephanopoulos)

Technical Skills

- **Bioprocessing & Fermentation:** Batch and fed-batch microbial fermentation, bioreactor operation with DO and pH control (BioFlo/CelliGen, DeltaV), scale-up, kLa and mass transfer, media and feed development, seed train expansion, tangential flow filtration (TFF), SOPs, batch records, tech transfer
- **Synthetic Biology & Metabolic Engineering:** Metabolic engineering, genome editing (Golden Gate assembly, CRISPR), plasmid design and cloning (*E. coli*), strain engineering, promoter library screening, adaptive laboratory evolution
- **Cell Culture & Molecular Biology:** Yeast, bacterial, and mammalian (CHO, Vero) cell culture, aseptic technique, cGMP, Vi-CELL, Cedex Bio, NucleoCounter, flow cytometry, qPCR, protein purification, Western blotting
- **Computational Biology & Data Science:** Python, R, SQL, MATLAB, JMP, statistical modeling, machine learning, data visualization, DoE, CFD (Ansys CFX)
- **Bioinformatics & Omics:** RNA-seq and Tag-seq (DESeq2), genomic variant analysis, genome annotation, metabolomics, high-performance computing, Docker
- **Analytical Chemistry:** HPLC, LC-MS/MS, GC-MS, UV-Vis, assay development, plate reader assays, microscopy
- **Software:** Benchling, LIMS, Linux, Bash, Git/GitHub, Jupyter, LaTeX, GraphPad Prism

Select Honors and Awards

- **First Place, Professional Division at the Data Challenge Expo, JSM (Joint Statistical Meetings)**, 2025
- **National Science Foundation Graduate Research Fellowship (NSF GRFP)**, 2021
- **Provost's Graduate Excellence Fellowship**, UT Austin, 2019
- **Individual Accomplishment Citation, MIT Department of Chemical Engineering**, 2018 & 2019
Only undergraduate to win this award two years in a row, for volunteer work and service to the department

Publications

14 peer-reviewed publications, seven as first or co-first author (* co-first). Selected:

- **Coleman, S. M.*** et al. High oil density bioreactor-scale fermentations of *Yarrowia lipolytica* using CFD modeling and experimental validation. *Biotechnology Journal* (2024)
- **Coleman, S. M.*** et al. Evolving tolerance of *Yarrowia lipolytica* to hydrothermal liquefaction aqueous phase waste. *Applied Microbiology and Biotechnology* (2023)

Full list: sarahmcoleman.github.io/papers.html